energywave.ai • Humanoid power-budget specification — the load the power supply must meet
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CONFIDENTIAL — DRAFT — FOR DISCUSSION · EVERY SUBSYSTEM LOAD IS AN INPUT, NOT AN ASSERTION

# The load, computed, not asserted. Three-month objective #2, as a worksheet.

The deck's claim is specific: a humanoid that runs human-grade sight and sound, smell and taste, touch and motion at once draws more than a lithium battery delivers. This specification turns that claim into numbers that can be checked. It defines the method, lists the published benchmarks with their sources, and leaves every subsystem load as an input to be measured or taken from a datasheet. No figure here is an EnergyWave measurement.

## 1 · METHOD

| Quantity | Definition |
|---|---|
| Average electrical load $\bar{P}$ | Σ over subsystems of (average power while active × duty cycle), plus housekeeping (compute, communications, thermal management, conversion losses). |
| Peak load $P_{peak}$ | The coincident maximum — actuators under load with sensing and compute active. Sets the supply's power ceiling and ramp rate. |
| Energy per shift E | $\bar{P}$ × shift length (default 8 h). The number that sizes a pack or a source. |
| Carried-mass equivalent m | E ÷ pack specific energy (Wh/kg). Compared with what platforms carry today and with the biological benchmark. |
| Gap | The ratio of E to the energy a platform-sized pack holds at today's pack specific energy; and the runtime a platform-sized pack would give at $\bar{P}$. |

## 2 · BENCHMARKS — EVERY FIGURE ATTRIBUTED, NONE MEASURED BY ENERGYWAVE

| Benchmark | Value | Source (status) |
|---|---|---|
| Human basal metabolic rate | 63–84 W (55-kg woman to 70-kg man) | FAO/WHO/UNU 2004 (reference data) |
| Human daily expenditure | 2.3–4.6 kWh/day (sedentary to vigorous) | FAO/WHO/UNU 2004 (reference data) |
| Humanoid power at rest | 150–400 W; 5–10× less efficient than a human in locomotion | Kashiri et al., Front. Robot. AI 2018 (peer-reviewed review) |
| Figure 03 | 2.3 kWh pack; 5 h at peak performance → ≈ 460 W average | Figure AI, "F.03 Battery Development", Jul 2025 (manufacturer claim) |
| Boston Dynamics Atlas (electric) | 4 h; 2 h with heavy lifting; 1.5 h charge | Atlas spec sheet, Dec 2025 (manufacturer claim) |
| Agility Digit | Up to 4 h; deployed cycles ≈ 90 min run / 9 min charge | Agility, Mar 2025; IEEE Spectrum, Sep 2025 (manufacturer claim; trade press) |

| Benchmark | Value | Source (status) |
|---|---|---|
| Unitree G1 · H1 | ≈ 0.43 kWh, ≈ 2 h · 0.864 kWh | unitree.com; reseller listing (manufacturer claim) |
| Tesla Optimus | 2.3 kWh, 52 V; “about a full day of work” | Tesla AI Day via IEEE Spectrum, Sep 2022 (manufacturer claim, dated) |
| Lithium-ion cell specific energy | ≈ 220 Wh/kg commercial (2020); 250–271 Wh/kg best cylindrical cells | U.S. DOE Battery500 update 2020; BatteryDesign.net benchmarks (reference data) |
| Pack specific energy | 120–200 Wh/kg; BEV packs ≈ 160–175 Wh/kg in 2026 benchmarks | BatteryDesign.net (reference data) |
| Body fat | 37 MJ/kg ≈ 10.3 kWh/kg; adipose tissue ≈ 8.9 kWh/kg | FAO Food and Nutrition Paper 77; Wishnofsky (reference data) |
| Battery improvement rate | ≈ 4 % per year industry planning assumption | cited in the August 2026 white paper from a patent filing (secondary) |

## 3 · THE WORKSHEET — SUBSYSTEM LOADS TO BE SUPPLIED

| Subsystem (deck slide 4) | Components | Average load W | Duty cycle | Basis to be recorded |
|---|---|---|---|---|
| Sight & sound | Cameras and microphones; associated vision and audio processing | **[Needs Input from Ian or Chip]** | **[Needs Input]** | Datasheet or measured draw of the chosen sensor set and the processing it requires. |
| Smell & taste | Gas-sensor and electrochemical arrays — electronic nose and tongue, including heaters where the sensor needs them | **[Needs Input from Ian or Chip]** | **[Needs Input]** | Datasheet; metal-oxide gas sensors are heater-dominated and must be counted at their heater draw. |
| Touch | Tactile arrays and electronic skin at nerve-level granularity; readout electronics | **[Needs Input from Ian or Chip]** | **[Needs Input]** | Per-taxel draw × count × scan rate; readout multiplexing. |
| Motion | Dense actuators at muscle-level granularity — body and hand; holding a position under gravity is a continuous draw | **[Needs Input from Ian or Chip]** | **[Needs Input]** | Actuator count × average electrical power by task profile; the dominant term on every published platform. |
| Compute and communications | Perception, planning, control, radios | **[Needs Input from Ian or Chip]** | ≈ 1 | Datasheet of the chosen compute module at the intended utilization. |
| Thermal management and conversion losses | Fans, pumps, DC–DC conversion | **[Needs Input from Ian or Chip]** | ≈ 1 | Conversion efficiency 80–90 % battery-to-motor (Kashiri 2018) sets the loss floor. |
| **Total average load P̄** | Σ of the rows above. **Anchor for sanity:** the only platform publishing both pack energy and runtime implies ≈ 460 W average at peak performance (Figure 03, manufacturer claim); humanoids draw 150–400 W at rest (Kashiri 2018). | | | |

## 4 · WORKED STRUCTURE — WITH THE ANCHOR FIGURE, NOT A COMPANY NUMBER

| Step | Using the Figure 03 anchor (≈ 460 W) | With EnergyWave's inputs |
|---|---|---|
| Energy per 8-h shift | ≈ 3.7 kWh | $\bar{P}$ × 8 h **[Needs Input: $\bar{P}$]** |
| Pack mass at 160 Wh/kg (pack level) | ≈ 23 kg | E ÷ 160 Wh/kg |
| Pack mass at 250 Wh/kg (cell level, no housing) | ≈ 15 kg | E ÷ 250 Wh/kg |
| Runtime of a 2.3-kWh pack | ≈ 5 h, as the manufacturer states | 2.3 kWh ÷ $\bar{P}$ |
| Biological comparison | A person's whole day is 2.3–4.6 kWh from fuel at ≈ 10 kWh/kg | E versus the human day; mass ratio at 10 kWh/kg versus 0.16 kWh/kg |

The anchor row shows the method on a published pair of numbers; it is not an EnergyWave finding and the platform's task profile is unknown. The deck's premise is that human-grade sensing and actuation push $\bar{P}$ above today's platforms; the worksheet is how that premise becomes a number with a source.

## 5 · SPECIFICATION OUTPUTS — WHAT THE POWER SUPPLY MUST MEET

| Requirement | Value | Derivation |
|---|---|---|
| Continuous electrical output | **[Needs Input from Ian or Chip: $\bar{P}$ from §3]** | Average load including housekeeping. |
| Peak output and ramp | **[Needs Input from Ian or Chip: $P_{peak}$, ramp rate]** | Coincident actuator peak; fixed in hardware under the control architecture (white paper §4). |
| Energy per shift | **[Needs Input: E]** | $\bar{P}$ × shift length **[Needs Input from Ian or Chip: shift length and duty profile]** |
| Mass and volume budget | **[Needs Input from Ian or Chip: target platform mass and bay volume]** | Set by the platform; today's packs occupy the torso. |
| Thermal limit | **[Needs Input from Ian or Chip: allowable heat rejection]** | Conversion losses at the stated efficiency. |
| Target platform | **[Needs Input from Ian or Chip: which humanoid, or a generic class]** | Every row above depends on it. |

## 6 · WHAT THIS SPECIFICATION IS NOT

It does not say that any EnergyWave device meets any row of §5, and it does not describe one. It states the load. Whether a non-chemical source can meet it is the research question the gates exist to answer, and the fourteen orders of magnitude between the only peer-reviewed unexplained output in the literature (1.4 pW, one laboratory, unreplicated) and the hundreds of watts in §4 are the honest measure of the distance.

Sources: Oct-13 deck slides 3–4 (S-01); Technical Prospectus v1 §§3.1–3.4 and Appendix A refs 32–36 (M-05); Investor Overview slides 3–4 (M-04); DoD white paper v2.0 §§1.1–1.3 (M-07); Physical Architecture & Baseline Model v1.3 (M-14) for the conversion-loss treatment.